

Janak Kabra

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

Vellore, Tamil Nadu

Aug. 2024 – May 2028

EXPERIENCE

AI Research Intern

Oct. 2025 – Dec. 2025

Mythus AI Studios

Remote

- Partnered with a cross-functional ML team to certify classification models for production, and cut pipeline runtime by ~20% by rewriting feature-selection scripts with vectorized NumPy and multiprocessing.

Software Intern

Feb. 2026 – Jun. 2026

Infosys Springboard

Remote

- Owned feature engineering and model selection within a project team, building LSTM and regression forecasting pipelines and driving adoption of the lowest out-of-sample RMSE specification.

Selected Builder (competitive cohort)

Swiggy Builders Club

Remote

- Selected into Swiggy's competitive builder program; built SnapOrder, a live multimodal AI ordering agent (Next.js, TypeScript, LLM-driven tool use) that turns a food photo into a placed order.

PROJECTS

raftkv: Linearizable Distributed KV Store | [GitHub](#) | [Live Demo](#) | [Go](#)

- Implemented the Raft consensus algorithm from scratch with no libraries (leader election, log replication, persistence, snapshotting, membership changes, ReadIndex linearizable reads), backing a fault-tolerant key-value store.
- Proved correctness with deterministic simulation testing: a seeded, Jepsen-style nemesis injects partitions and crashes while a from-scratch Wing-Gong-Lowe checker verifies linearizability; every failure replays from its seed, all under the Go race detector.
- Benchmarked consensus cost on a 3-node cluster (linearizable read $\approx 143\mu s$, local stale read $\approx 71\mu s$, write $\approx 1.2ms$) and shipped a live 5-node visualizer with real-time leader-failover.

Low-Latency Query Engine | [GitHub](#) | [Live Demo](#) | [FastAPI](#), [PostgreSQL](#), [React](#)

- Cut multi-factor query latency to under 500ms over a 500-row universe by designing targeted indexes and collapsing chained filters into a single database round-trip.
- Delivered real-time interactive screening through a REST API and React UI spanning valuation, profitability, and leverage factors.

High-Performance Numerical Engine | [GitHub](#) | [Live Demo](#) | [Python](#), [Numba JIT](#), [FastAPI](#), [Docker](#)

- Built a pricing and calibration engine from first principles and JIT-compiled the hot kernels with Numba, bringing evaluation into the sub-millisecond range.
- Diagnosed and fixed a numerical-overflow instability at long horizons by restructuring the kernel into a stable branch-cut-free form, making previously-divergent inputs computable.
- Shipped behind a Dockerized FastAPI service with GitHub Actions CI and a 122-test suite, plus a live interactive demo.

Algorithm Execution Debugger | [GitHub](#) | [Live Demo](#) | [React](#), [TypeScript](#), [Canvas](#)

- Made abstract Big-O behavior directly observable with a 60-FPS Canvas visualizer instrumenting Quick, Heap, and Merge sort by live comparison and swap counts.
- Engineered a zero-allocation step recorder producing immutable snapshots, enabling deterministic replay and step-through debugging.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript/JavaScript, SQL, C++, C, Java

Backend & Systems: Distributed Systems, Consensus (Raft), REST API design, FastAPI, Node.js, PostgreSQL, Docker, Linux, Git, GitHub Actions (CI/CD)

Frontend: React, Next.js, Tailwind CSS

Libraries & Tools: NumPy, Pandas, SciPy, Numba JIT, Plotly, Postman

CS Foundations: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Concurrency